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The F-35's vertical landing and hover capability was demanded by the US Marine Corps

F-35: A Trillion Dollar Mistake

"A camel is a horse designed by a committee" goes a proverb that applies all too well to the horribly overbudget and underperforming F-35 project.

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The decline of Russia following the end of Cold War meant that it couldn't keep up with the US pursuit of achieving absolute air dominance. It's little wonder why the stealth-capable Lockheed Martin F-22 Raptor asserted its dominance as the world's premier fighter jet platform largely unchallenged at the turn of the millennium. Nothing underscores this better than the fact that Russia's fifth generation answer to the US stealth fighter, PAK-FA, still hasn't been deployed.

The Lockheed Martin F-35 Lightning II is the successor to the mighty F-22 Raptor. Prima facie, that makes it sound like something that would destroy everything in the airspace and then fly over to Mars to take on the space worms. And that seems pretty accurate if you go by the sheer amount of destruction it has wrought on

the silver screen. However, the Hollywood versions of the upcoming fighter jet invariably end up in a spectacular ball of flame after destroying half the city in movies such as The Avengers, Man of Steel, Green Lantern or Die Hard 4.0. This fate just might ring true for its real-life counterparts but more on that later.

Multirole Fighter

Although the F-35 is supposed to be a successor to the F-22, it hasn't exactly been designed to surpass its predecessor in terms of performance. At a unit cost of \$361 million, the F-22 was a silver bullet commissioned at the height of the Cold War to deny Russians access to air superiority. Let me reiterate, that's the price of one piece and not the entire F-22 project. The end of Cold War and the decline of the Russian war machinery meant that the United States Air Force (USAF) was in the market for a more cost effective fifth generation fighter. The F-35 programme is supposed

to replace existing fourth generation air-force and navy platforms such as the F-16 Fighting Falcon, F-18 Hornet, A-10 Thunderbolt II, and the AV-8B Harrier II.

Designed as a multirole fighter, the F-35 achieves this versatility through three separate platforms within the same family that share up to 80-percent parts for logistical simplicity and to ensure cost effectiveness. These platforms, classified according to takeoff and landing capabilities, include the F-35A (CTOL: Conventional Takeoff and Landing), F-35B (STOVL: Short Takeoff and Vertical Landing), and F-35C (carrier-based version). The F-35A variant is designed according to USAF needs and is the closest to a regular fighter jet design, which does away with the complicated and weight adding STOVL capability. The F-35B was designed to fulfill the US Marine Corps requirement of short takeoff and vertical landing, which adds to the total weight and fuel requirements for the difficult to achieve (and control) hovering and vertical